Proceedings of the International Conference on Natural Language Processing & Engineering · Published 2024

Hybrid Lexical and Dense Vector Indexing for Fast Domain-Specific Academic Question Answering

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ABSTRACT

Pure dense vector search struggles with specialized academic queries containing exact mathematical symbols, chemical formulas, and dataset names. We formulate a reciprocal rank fusion hybrid indexing architecture that dynamically balances sparse BM25 lexical inverted indices with dense 768-dim embeddings. Our system computes dynamic query entropy to adjust lexical-versus-semantic weighting: queries with exact acronyms bias toward lexical search, while conceptual questions favor embeddings. Across academic question-answering benchmarks, our hybrid pipeline achieves 93.6% top-5 retrieval accuracy with a sub-12ms response latency. The primary limitation is vocabulary mismatch when newly coined scientific terms are absent from tokenizers and lexical frequency tables.

Keywords— Natural Language Processing, Hybrid Search, BM25, Vector Search

INTRODUCTION

Scientific researchers search using precise entity identifiers ('ResNet-50', 'Cas9-sgRNA') where dense embeddings fail due to token splitting, yet also search using broad conceptual queries ('methods to mitigate catastrophic forgetting') where keywords fail.

METHODOLOGY

We implement a dual-stage retriever combining an SQLite FTS5 lexical index with a FAISS vector index. A classifier inspects query token distributions and computes optimal linear combination weights for Reciprocal Rank Fusion (RRF).

RESULTS

The hybrid retriever improved MRR@10 from 0.71 (pure BM25) and 0.74 (pure vector) to 0.89 across 50,000 scholarly queries, achieving sub-12ms query execution times.

LIMITATIONS

When encountering newly coined jargon with zero historical lexical frequency and sub-optimal tokenization, neither sparse nor dense branches achieve high confidence scores.

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